

AI Applications Lecture 4

Licenses and Openness

SUZUKI, Atsushi^a

^aI thank my collaborator, Jing WANG, for cooperating in creating the teaching materials.

Outline

1. Introduction
2. What Does "Open" Mean?
3. What rights do we have on open models? - License
4. Model Versions and License Differences Across Versions
5. Summary
6. Next Time

Review of the previous lecture

In the last lecture, we defined the function represented by a neural network as **generally as possible**.

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- A **checkpoint** contains the **specific parameter values** obtained through training.
- Only with both the architecture $f_{(.)}$ and the checkpoint θ is a **specific function** f_{θ} uniquely determined.

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For an input space $\mathcal{X}$, an output space $\mathcal{Y}$, and a parameter space $\Theta \subset \mathbb{R}^D$:

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- **Training** is determining parameters θ^* based on data.
- **Inference** is applying the final function f_{θ^*} .

From closed AI to open AI

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- **Completely open:** Architecture, checkpoints, and even the training code or "recipes" are released.
- A model's **license** determines if others can freely use it, regardless of how "open" it is.

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Example (The Staged Release of GPT-2)

When OpenAI's GPT-2 was announced in 2019, the weights for the largest model were subject to a **staged release** due to safety concerns [1].

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When OpenAI's GPT-2 was announced in 2019, the weights for the largest model were subject to a **staged release** due to safety concerns [1].

- Initially, the most powerful 1.5B parameter model was completely closed.
- The full model was eventually released in November 2019 after a period of study and debate [2].

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Example (LLMs as a Service)

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Example (LLMs as a Service)

As of August 2026, OpenAI provides **ChatGPT-5.6** via its ChatGPT interface and API.

- **Inference is executed on their servers.**
- The **details of the architecture and checkpoints are generally not disclosed** [3].
- Users only interact with prompts and outputs.

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Example ("Open" Models and Distribution Platforms)

Models like **Llama** (Meta), **Qwen** (Alibaba), **Mistral** (Mistral AI), and **DeepSeek** (DeepSeek AI) are known as **open (Open Weights/Open Model)**.

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- The **Hugging Face Hub** is a major platform for their distribution [4].
- You can download the architecture and checkpoints to run locally.
- Licenses are (usually) clearly stated in the model repositories. [5, 6, 7, 8, 9]

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By the end of this lecture, students will be able to:

- Explain the various stages of openness of neural network models.
- Avoid the risk of not complying with the license.
- Determine whether a model provided under an major open source software (OSS) license can be used for their own purposes.

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- The goal is to get aware of basic legal risks.
- **Final decisions should be made at your own responsibility. Refer to official documentation [10, 11, 12].**

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Question: When we say an AI model is open, what does exactly mean?

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- The **architecture** (as configuration files or code).
- The trained **checkpoint** (the weights).

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- The **training data**.
- The **training algorithm** (including the complete training recipe and scripts for reproduction).

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With the architecture $f_{(\cdot)}$ and checkpoint θ , **inference** is possible.

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However, **reproducing the training** (arriving at the exact same θ) is usually impossible unless the data, recipe, random seeds, etc., are identical.

Even with "open weights," if the **insights from the training process** are not disclosed, the reproducibility and potential for adaptation are limited.

Closed models

UI/API-only Release

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- The **weights and detailed implementation remain private**.
- Example: OpenAI's generally available models used in ChatGPT4 and beyond [3].
 - Note: OpenAI also offers open weight models¹

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- Examples: Mistral 7B (Apache 2.0) [7], DeepSeek-R1 (MIT) [8].

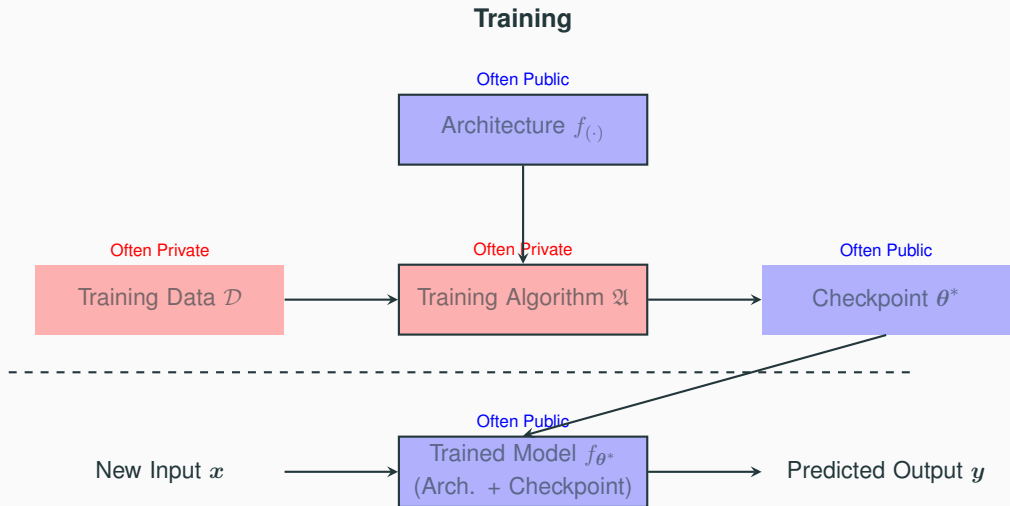
“Typical” open-weight models

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- Weights are distributed as **Open Weights**.
- This allows for local execution or use as a starting point for retraining.
- Examples: Mistral 7B (Apache 2.0) [7], DeepSeek-R1 (MIT) [8].
- The **training data and complete training procedures** are often not disclosed.

“Typical” open-weight models

What’s open vs. private for a typical open-weight model?



“Completely” open model

Release of Training Code and Recipes

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- Example: AI2’s OLMo project [13].

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Question: When we say an AI model is open, what does exactly mean?

Answer: In most cases, it means that its checkpoint (weights) is open (publicly available).

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Hence, even if an AI model is open, using it without permission may have a legal risk.

The distributor of an open AI model may provide permission through a **license**. Hence, understanding licenses is necessary when using open AI models.

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For example, within the same Transformers framework:

- **DeepSeek-R1** weights are MIT [8].
- The **Mistral** family is Apache 2.0 [7].
- The **Llama** family has its own custom license [5, 6].

Architecture vs. checkpoint licenses

Moreover, different licenses may apply to different components. Specifically, the license for the **code** and the license for the **weights** can be different.

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According to the model card, the DeepSeek-R1 checkpoint is licensed under the MIT license [8], even though we can use it via the Hugging Face Transformers library, which is licensed under the Apache 2.0 license [14].

Hence, it is necessary to read licenses for each component of the AI model you want to use.

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Hence, it is necessary to read licenses for each component of the AI model you want to use.

Fortunately, many famous open models are licensed under a major open source software license, so it is beneficial to know those licenses in advance.

Is a checkpoint protected by copyright?

Remark

In most regions of the world, court decisions are limited in whether a checkpoint is protected by copyright. In any case, it is safer to use an AI model only when the license gives explicit permission.

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Remark (Disclaimer)

Final determination of "can I use this?" must be based on the **specific license text and its scope**.

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- **Model examples:** [DeepSeek-v3](#), [DeepSeek-R1](#) [8].

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- **Model/Library example:** The classic machine learning library **scikit-learn** [16].

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- **Model Example**: **YOLOv5** uses AGPL-3.0 [19].

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- Careful consideration is required, especially for commercial use.

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Model-specific licenses (open-weight type)

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Recent **open-weight** models sometimes use **custom terms that are not standard OSS licenses**.

These often include use restrictions or thresholds based on Monthly Active Users (MAU).

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The **Community License** for the Llama 3 series allows **commercial use with restrictions** [5, 20].

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- Later versions, like Qwen 3, seem to have standardized on Apache 2.0.

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It also requires that when distributing **derivative works**, the **same use restrictions must be imposed on downstream users**.

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- Key variations include:
 - **CC BY 4.0** (Attribution) [22]
 - **CC BY-SA 4.0** (ShareAlike) [23]
 - **CC BY-NC 4.0** (NonCommercial) [24]

Practical guidelines and caveats

- Seek legal experts' advice.
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Practical guidelines and caveats

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- A policy of **"when in doubt, do not use"** is the safest approach.
- Seek legal experts' advice.

License - answer

Question: Under which conditions can we use open AI models?

Answer: Users' rights are limited owing to patents, copyright, contracts, etc. The distributor may give permission through its license.

Outline

1. Introduction

2. What Does "Open" Mean?

3. What rights do we have on open models? - License

4. Model Versions and License Differences Across Versions

5. Summary

6. Next Time

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Example (Qwen 2.5 (Alibaba))

As we saw, different models in the Qwen 2.5 series had different licenses (custom vs. Apache 2.0) [6]. You must check the license for the specific model you wish to use.

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- **Scale:** The number of parameters (e.g., 7B vs 72B).
- **Fine-tuning:** Whether it's a base model or tuned for a specific task (e.g., "Instruct").

Number of parameters (billion parameters)

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In mathematical terms, the parameters (checkpoint) are a real vector:

$$\theta = (\theta_1, \theta_2, \dots, \theta_D) \in \mathbb{R}^D.$$

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Generally, a higher parameter count increases expressive power but also computational cost.

Numerical representation

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- **Quantization**: Using even fewer bits (e.g., **int8**, **int4**). Drastically reduces size and speeds up inference, often with a small accuracy drop.

Precise definition of floating-point numbers

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Table 1: Bit Allocation for Floating-Point Formats

Format	Total Bits	Sign	Exponent	Mantissa	Bias
fp32	32	1	8	23	127
fp16	16	1	5	10	15
bf16	16	1	8	7	127

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Example (A number representable in bf16 but not in fp16)

Large number: Consider $V = 100,000$.

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Large number: Consider $V = 100,000$.

- **fp16:** The maximum representable value is 65,504. Therefore, 100,000 **overflows** to infinity (∞).
- **bf16:** Has the same dynamic range as fp32. It can be **represented without issue**.

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A number requiring precision: Consider $V = 1 + 2^{-8} = 1.00390625$. In binary, this is $(1.00000001)_2$.

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- **fp16:** The mantissa has 10 bits. It can **accurately represent** the '1' in the 8th decimal place.
- **bf16:** The mantissa has only 7 bits. The 8th bit does not fit and is rounded. It **cannot distinguish** between 1 and $1 + 2^{-8}$.

Instruction finetuning

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- Additional fine-tuning is performed using a dataset of "instruction (prompt) and desired response" pairs [28].
- This teaches the model to be a helpful assistant, follow diverse instructions, and engage in conversation.

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- Even if a model is "public," whether **third parties can freely use it** is determined by its **license**.
- In practice, one must understand the nature of licenses like **MIT / BSD-3 / Apache 2.0 / GPL-3** and always check the **LICENSE and model card for each model**.

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- The **surrounding components** (tokenizer, pre/post-processing, inference runtime).

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